Number of Provinces (medium)

Problem Statement

There are `n` cities. Some of them are connected in a network. If City `A` is directly connected to City `B`, and City `B` is directly connected to City `C`, city `A` is directly connected to City `C`.

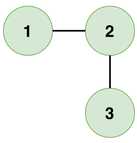
If a group of cities are connected directly or indirectly, they form a **province**.

You are given a square matrix of size `n x n`, where each cell's value indicates whether a direct connection between cities exists (1 for connected and 0 for not connected).

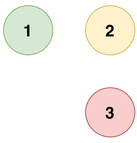
Determine the total number of provinces.

Examples

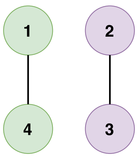
- **Example 1**:
  - Input: `[[1,1,0],[1,1,1],[0,1,1]]`



- Expected Output: `1`
- Justification: Here, all cities are connected either directly or indirectly, forming one province.
- **Example 2**:
  - Input: `[[1,0,0],[0,1,0],[0,0,1]]`



- Expected Output: `3`
- Justification: In this scenario, no cities are connected to each other, so each city forms its own province.
- **Example 3**:
  - Input: `[[1,0,0,1],[0,1,1,0],[0,1,1,0],[1,0,0,1]]`



- Expected Output: `2`
- Justification: Cities 1 and 4 form a province, and cities 2 and 3 form another province, resulting in a total of 2 provinces.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    private int[] parent; // Array to store the root of each node.

    // Function to find the province count.
    public int findProvinces(int[][] isConnected) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}
```

Test Case | Run Code Result | Stdout

```
[[1,1,0],[1,1,0],[0,0,1]]
```

Use Example Testcases | Run | Submit